

Conditions for HWE:

- 1) No mutations (Δ DNA)
- 2) No Nat select (VHDs)
- 3) No gene flow (migration)
- 4) No genetic drift (random Δ f(alleles))
- 5) Random mating (no bias / preference)

Study Questions for Lectures on Hardy-Weinberg Equilibrium

This study guide provides examples for the types of questions that you might see on the first term from the lectures over Hardy-Weinberg Equilibrium and the effects of violating Hardy-Weinberg assumptions. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture and the lecture slides and audio on Canvas to know what all topics have been covered.

1. True or False: If the genotype frequencies of a given population add up to 1, then that population is in Hardy Weinberg Equilibrium.

- a. False No, it must follow $p^2 + 2pq + q^2$
- b. True

2. The table below shows the observed genotype frequencies in a population along with the expected genotype frequencies under Hardy-Weinberg equilibrium.

Which of the following can you conclude from this data?

Genotype	Expected	Observed
AA	0.09	0.02
Aa	0.42	0.64
aa	0.49	0.34

- a. The population follows all Hardy-Weinberg assumptions False: Expected = Observed
- b. Inbreeding is likely occurring: Inbreeding yields high homozygous but hetero is winner here...
- c. There is probably selection in favor of the heterozygote
- d. Drift is likely high in this population
D = causes random fluctuations (doesn't produce clean patterns) But we see favor in heterozygous

3. Which of the following **is not** a violation of Hardy-Weinberg assumptions?

- a. Lizards preferentially choose mates with brighter dewlaps bias mate
- b. There are ten different alleles for a particular gene
- c. Foxes with tamer personalities are more likely to reproduce bias mate
- d. A new allele arises in the population drift
- e. A group of five bats colonize a new, previously uninhabited island migration

4. A gene for flower color in pea plants has two alleles, A and a. Given the following genotype frequencies, which of these populations shows evidence of **possible inbreeding**?

- Not U a. AA=0.01, Aa=0.30, aa=0.69 (Candidates: Low hetero (Aa), high homos (both AA and aa) "U distribution")
- Not U b. AA=0.04, Aa=0.32, aa=0.64 Find p, q:
- HWE c. AA=0.25, Aa=0.5, aa=0.25 Under HWE (Expected)
- d. AA=0.20, Aa=0.39, aa=0.41 $p(A) = AA + \frac{1}{2} Aa$
 $= 0.2 + 0.195 = 0.395 = p$
 $q = 1 - p = 0.605$
- e. AA=0.09, Aa=0.42, aa=0.49
- Under HWE (Expected)
- AA = $p^2 = 0.156 < .2$
Aa = $2pq = 0.478 > .39$
aa = $q^2 = 0.366 < .41$

5. In many large mammal populations, highways can be barriers to movement - individuals rarely cross them. One solution to this problem has been to install overhead bridges for wildlife to safely cross. The goal is to increase gene flow between the populations on either side of the highway. Which of the following would you expect to see if the bridge is successful in its goal?

- a. Higher levels of drift would occur in both populations after the bridge is installed Gene flow increase pop size making drift less likely
- b. The two populations would have different mutation rates mutations are random.
- c. The two populations would become more genetically similar
- d. Both populations would decrease in size why?
- e. More inbreeding would occur in both populations

Drift: (Diverges pops, diff)

- random f(allele) Δ , not because allele is better.
- Bad: Next gen could lose an important allele
 - introduce bad alleles
 - reduce genetic div
- Strong in small iso pops.

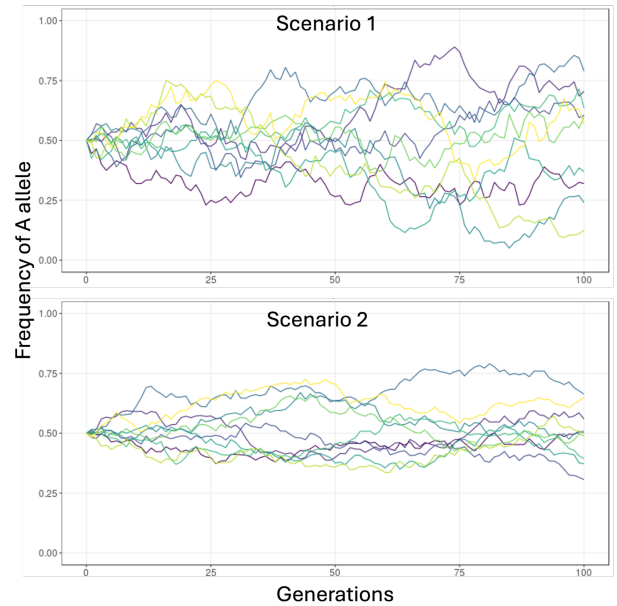
Gene Flow: (homogenize pop, same)

- Migrating + successfully reproducing w/ another population.

Drift vs Flow

6. We conducted computer simulations under two different scenarios, shown in the two figures below. Each line represents a separate population simulated under that scenario. The figure below shows how the frequency of the A_1 allele (y axis) changes over time (x axis). Which of the following can you conclude based on this data (select all that apply)?

- (a) Scenario 1 represents a smaller population size than Scenario 2 $SI = \text{more } f(\text{allele}) = \text{drift} = \text{small pop}$
- b. The phenotype produced by the A allele is beneficial in Scenario 1
- c. A new mutation arose in this population in Scenario 1
- d. The phenotype produced by the A allele is beneficial in Scenario 2



7. You are studying a population of grasshoppers and you find four different alleles at the A gene, with frequencies shown in the table. Assuming the population is in Hardy-Weinberg equilibrium, the frequency of genotype A_1A_3 is $2pr$ and the frequency for genotype A_2A_2 is q^2 .

$p^2 + 2pq + q^2$

Allele	Frequency
A_1 AA	p
A_2 Aa	q
A_3 aA	r
A_4 aa	s

8. A small population of 26 individuals has five alleles, A, B, C, D, and E for a particular gene. Five individuals are DA heterozygotes, five individuals are AA homozygotes, five individuals are AB heterozygotes, five individuals are CD heterozygotes, five individuals are CC homozygotes, and one individual is an EE homozygote.

- a. What is the frequency of each allele in this population?

Total = 52 Alleles

$A: (5 + 10 + 5) / 52$, $B: 5 / 52$, $C: (5 + 10) / 52$, $D: (5 + 5) / 52$, $E: (2 + 5) / 52$

0.3846 0.0962 ... just counting..

- b. If five AE heterozygotes migrate into the population, what will be the resulting allele frequencies?

9. A population of otters in H-W equilibrium has two alleles for fur color: D is dark brown and complete dominant to d, which is light brown. If 81% of the population has light brown fur, what is the frequency of homozygous dominant otters? $dd = \text{light brown} = .81 = q^2$

$\sqrt{q^2} = .9 \dots p = 1 - 0.9 = 0.1 \quad p^2 = 0.1 = 0.01$

10. A population of plants in H-W equilibrium has two alleles (B and b) for flower color. The frequency of the B allele in the population is 0.7. What is the frequency of plants in this population that are heterozygous for flower color? $BB = p = 0.7 \quad 2pq = ? \quad B \text{ allele} = \text{allele freq} = p \text{ or } q$

$q = 1 - 0.7 = .3 \quad 2pq = 2(.7)(.3) = 0.42$

allele = pq

11. A nearby population of snails in H-W equilibrium has two alleles (R and r) for shell color. If the frequency of snails that are homozygous for R is 0.64, what is the frequency of snails that are homozygous for r?

$R = p^2 = 0.64, p = 0.8 \quad q = 1 - 0.8 = 0.2 \quad R \text{ genotype} = \text{geno freq} = p^2 \text{ or } q^2$

geno = p^2q^2

$rr = q^2 = 0.2^2 = 0.04$

$$2pq = 0.42 \quad q = 1 - p \quad p(1 - p) = 0.21 \quad p - p^2 = 0.21 \rightarrow p^2 - p + 0.21 = 0 = (p - 0.3)(p - 0.7) = 0$$

$$p = 0.3 \text{ or } 0.7$$

12. A population of trees in H-W equilibrium has two alleles (F and f) for seed color. The frequency of trees heterozygous for seed color is 0.42. You know that the f allele is more common than the F allele in this population. What are the frequencies of each allele in this population? $f > F$ so $p = 0.3, q = 0.7$

13. A population of geckos in H-W equilibrium has two alleles for tail length: L for long tails, which is incomplete dominant to l for short tails. The frequency of geckos with long tails is 0.36. What percent of the population would you expect to be heterozygous? Given: $L = p^2 = 0.36, p = .6, q = 0.4, \frac{2(.6)(.4)}{1} = 0.48$

Phenotype: $p^2 + q^2 + r^2$

14. A population of bison in H-W equilibrium has three alleles for horn shape: long (A), short (a) and curly (α). The long allele is dominant over both the short and curly alleles, and the short allele is dominant over the curly allele. The frequency of the a allele is 0.5, and 16% of the population has curly horns.
- a. What percent of the population has long horns? Hint: remember that long horns is a phenotype that represents multiple genotypes $(.1)^2 + 2(.1)(.05) + 2(.1)(.4) = 0.19$ Trait question, not genotype.
- b. Which of the three phenotypes (long, short, curly) is the most common in this population?
- c. What is the frequency of homozygotes in this population? b) Short: $1 - .19 - .16 = .65$ 65% is short.
19% long
16% curly
- c) $p^2 + q^2 + r^2 = .01 + .25 + .16 = .42 = 42\%$

15. Given the following genotype frequencies, is each of these populations in Hardy-Weinberg equilibrium? If not, determine which observed genotype frequencies are greater or less than expected and propose an explanation for the observed phenotypes (ie. Which H-W assumption is violated?).

Note: there may be more than one possible explanation for the observed genotype frequencies

- (a) $AA = 0.01, Aa = 0.18, aa = 0.81$ $f(A) = 0.1 + 0.5(.18) = 0.1 = p$
 (b) $AA = 0.25, Aa = 0.50, aa = 0.25$ $f(a) = 1 - f(A) = .9 = q$
 (c) $AA = 0.36, Aa = 0.60, aa = 0.04$ $f(A_{exp}) = 0.1^2 = 0.01 \checkmark, f(Aa_{exp}) = 2(.1)(.9) \checkmark$
 (d) $AA = 0.09, Aa = 0.42, aa = 0.49$ $f(a_{exp}) = 0.9^2 = 0.81 \checkmark$
 (e) $AA = 0.64, Aa = 0.20, aa = 0.16$ Do this for remaining ↑ ...
- From glance: Aa too big
- Small group migration

16. How are the founder effect and a population bottleneck similar? How are they different? Which one would you predict to have a more serious negative impact on the species as a whole?

Same: Reduce genetic Diversity by reducing population size | Diff: OG pop for founders still exist and maintains genetic div in that area
 Bottleneck pop suffers more genetically.

17. Describe the consequences of genetic drift. Why are small populations more affected by drift than larger populations? Smaller pops suffer more from random sampling effects. Ea indiv reprs a larger portion of gene pool so chance event (who mates who) cause larger swings in allele frequencies like loss or total allele fixation.
18. Gene flow and genetic drift are sometimes referred to as 'non-adaptive evolution.' What does this phrase mean? What type(s) of evolution is/are adaptive? They considered non-adaptive bc they change allele frequencies ignoring fitness. Gene flow is due to population changes from movement while drift comes the small sampling effect.
19. Describe the genetic mechanism by which inbreeding increases homozygosity in a population.

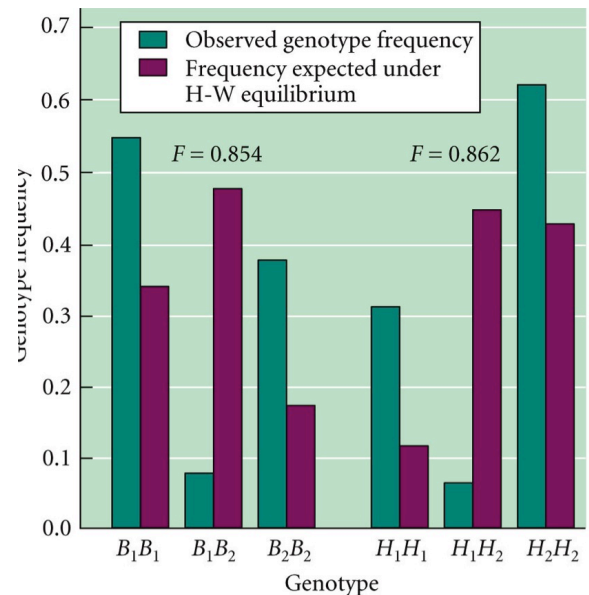
- a. If a population of plants begins with two alleles being equally common ($p=0.5$ and $q=0.5$) and all plants self-fertilize, what would you expect the frequency of heterozygotes to be in the first generation? In the 50th? In the 100th? Hint: there's a formula for this – look for a pattern in the first few generations. $(0.5)^1, (0.5)^{50} = 8.88e-16, (0.5)^{100} = 7.99e-31$
- b. After how many generations would you predict there will be no heterozygotes in the population? Mathematically we can never hit 0.
 But after enough gens it becomes so small that'd we'd to not see any hetero indivs.

20. We listed five conditions that a population must meet in order to be in Hardy-Weinberg Equilibrium. How might violation of one assumption lead to or be affected by violations of other assumptions? For example, how might natural selection affect the fate of a new mutation in a population?
If new mutated allele is beneficial natural selection will cause it to increase in population. Or vice versa if its doing harm to fitness.

21. The figure at right shows the observed and expected genotype frequencies for a population of *Avena fatua* grass.

- Is this population in Hardy-Weinberg Equilibrium? If not, describe what assumptions of Hardy-Weinberg Equilibrium could be violated in this population. *Hint: there is more than one possible answer*
NO. Observed & Expected aren't equal
- Are the two genes experiencing the same pattern? How does your answer to that question narrow down the possible options you listed in part A?

*Homo > hetero for both traits B, H
 Suggests inbreeding.*



22. The table below shows the genotype frequencies for a particular gene observed in a fictional population of otters, and the expected genotype frequencies if this population was in Hardy-Weinberg Equilibrium. What violations of HWE conditions could be affecting this population? *Select all that apply*

	RR	Rr	rr
Observed Frequencies	0.43	0.21	0.36
Expected Frequencies under HWE	0.15	0.19	0.66

Compare : RR much higher expected Rr expected much lower than expected

- ☒ Migration of rr individuals out of the population
- ☒ Genetic drift increasing the frequency of the R allele
- ☐ Inbreeding *rr too low for inbreeding*
- ☒ Directional selection in favor of the RR phenotype
- ☐ Mutation of R alleles into r alleles
We have less r, not more. This is contradictory.